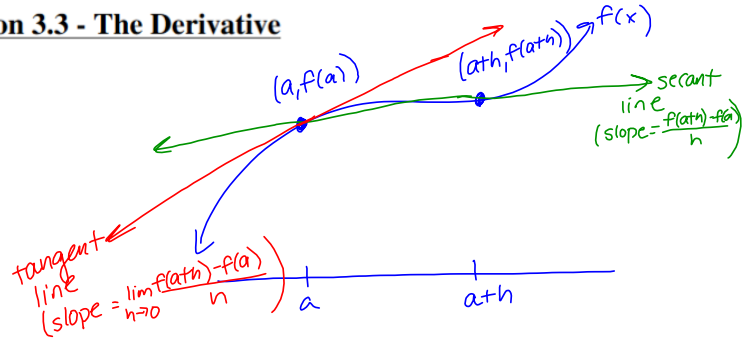
## Section 3.3 - The Derivative




**Definition:** The **derivative of a function** $f$ **at a number** $a$, denoted by $f'(a)$, is

$$\boxed{f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}}$$

provided that this limit exists.

### Interpretations of the Derivative

The derivative has various applications and interpretations, including the following:

1. **Slope of the tangent line:** $f'(a)$ is the slope of the line tangent to the graph of $f$ at the point $(a, f(a))$.
2. **Instantaneous rate of change:** $f'(a)$ is the instantaneous rate of change of $y = f(x)$ at $x = a$.
3. **Velocity:** If $f(x)$ is the position of a moving object at time $x$, then $v = f'(a)$ is the velocity of the object at time $x = a$.

**Example 1:** Given $f(x) = 2x^2 + 8$, find the following:

a) $f'(-2)$.

$$f'(-2) = \lim_{h \to 0} \frac{f(-2+h) - f(-2)}{h}$$
$$= \lim_{h \to 0} \frac{\cancel{16} - 8h + 2h^2 - \cancel{16}}{h}$$
$$= \lim_{h \to 0} \frac{-8h + 2h^2}{h}$$
$$= \lim_{h \to 0} \frac{\cancel{h}(-8+2h)}{\cancel{h}}$$
$$= \lim_{h \to 0} (-8+2h) = -8 + 2(0) = \boxed{-8}$$

$f(-2+h)$
$= 2(-2+h)^2 + 8$
$= 2(-2+h)(-2+h) + 8$
$= 2(4 - 4h + h^2) + 8$
$= 8 - 8h + 2h^2 + 8$
$= 16 - 8h + 2h^2$

$f(-2) = 2(-2)^2 + 8 = 16$

b) The equation of the tangent line at $x = -2$

$(-2, f(-2))$ $(-2, 16)$ $m = -8$

$y = mx + b$
$16 = -8(-2) + b$
$16 = 16 + b$
$0 = b$

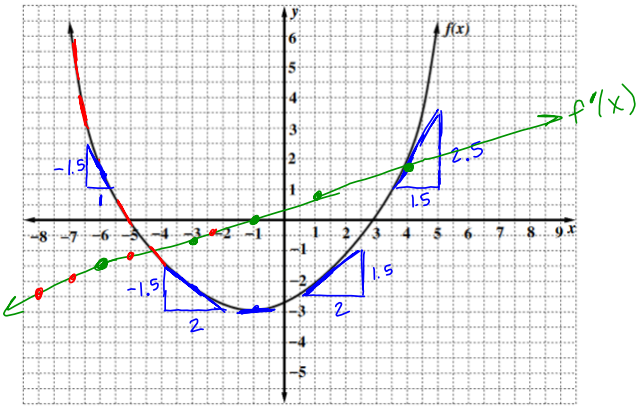
$\boxed{y = -8x}$

If the number $a$ is allowed to vary in our definition from the previous page, we obtain a new function called the derivative of $f$, $f'(x)$, which is defined as

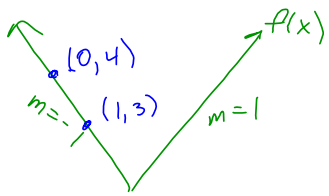
$$\boxed{f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}}$$

Other Notations for the Derivative of $y = f(x)$: $f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x)$

**Example 2:** Given the graph of $f(x)$ below, estimate the value of $f'(x)$ at the following points:




| $x$ | $-6$ | $-3$ | $-1$ | $1$ | $4$ |
|---|---|---|---|---|---|
| $f'(x)$ | $\frac{-1.5}{1} = -1.5$ | $\frac{-1.5}{2} = -.75$ | $0$ | $\frac{1.5}{2} = .75$ | $\frac{2.5}{1.5} \approx 1.67$ |




$m = \frac{4-3}{0-1}$

**Conclusion:**

- when $f(x)$ is increasing: $f'(x) > 0$
  (lies above the x-axis)
- when $f(x)$ is decreasing: $f'(x) < 0$
  (lies below the x-axis)
- when $f(x)$ is constant: $f'(x) = 0$
  (touches the x-axis)

**Example 3:** Find $f'(x)$ if $f(x) = \dfrac{1-x}{2+x}$.

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h\to 0} \frac{\frac{1-(x+h)}{2+x+h} - \frac{1-x}{2+x}}{h}$$

$$= \lim_{h\to 0} \frac{1}{h}\left(\frac{(2+x)(1-x-h)}{(2+x)(2+x+h)} - \frac{(1-x)(2+x+h)}{(2+x)(2+x+h)}\right)$$

$$= \lim_{h\to 0} \frac{1}{h}\left(\frac{\cancel{2}\cancel{-2x}-2h+\cancel{x}\cancel{-x^2}\cancel{-xh}-(\cancel{2}+\cancel{x}+h\cancel{-2x}\cancel{-x^2}\cancel{-xh})}{(2+x)(2+x+h)}\right)$$

$$= \lim_{h\to 0} \frac{1}{h}\left(\frac{-3h}{(2+x)(2+x+h)}\right)$$

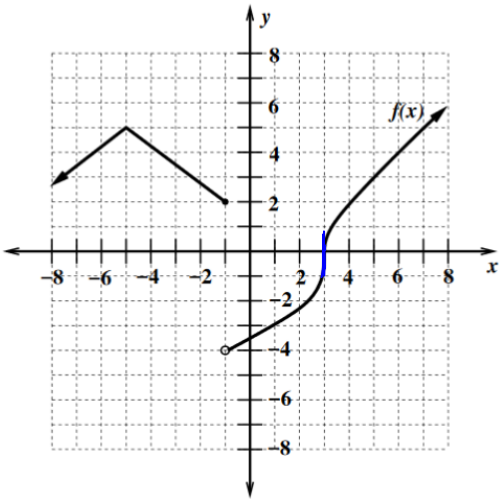
$$= \lim_{h\to 0} \frac{-3\cancel{h}}{\cancel{h}(2+x)(2+x+h)} = \lim_{h\to 0} \frac{-3}{(2+x)(2+x+h)} = \frac{-3}{(2+x)(2+x+0)} = \frac{-3}{(2+x)(2+x)} = \boxed{\frac{-3}{(2+x)^2} = f'(x)}$$

**Nonexistence of the Derivative**

If $f'(a)$ does not exist, then we say that $f(x)$ is **nondifferentiable** at $x = a$. This occurs when the graph:

1. has a sharp turn or corner
2. has a discontinuity
3. has a vertical tangent line

**Example 4:** For what values of $x$ is $f$ nondifferentiable?

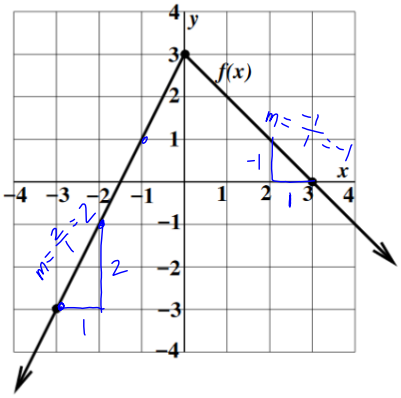


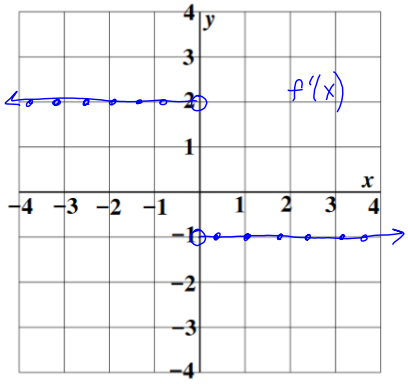

$x=3$ (vertical tangent)

$x=-1$ (discontinuity)

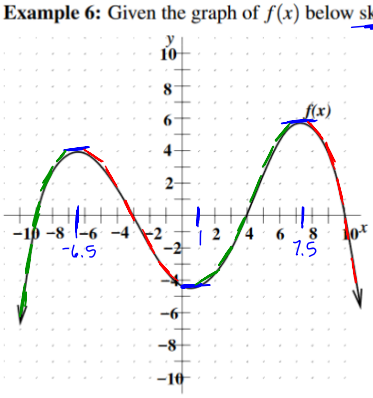
$x=-5$ (sharp turn)

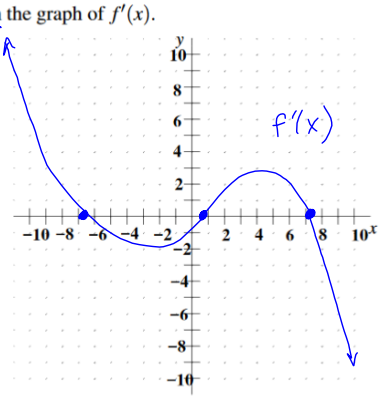
**Example 5:** Given the graph of $f(x)$ below sketch the graph of $f'(x)$.




**Example 6:** Given the graph of $f(x)$ below sketch the graph of $f'(x)$.




**Section 3.3 Highly Suggested Homework Problems: 1-13(odd), 23-33(odd), 37-43(odd), 47, 53, 55**